

— SAP ADVISORY ARCHITECTURE

# Archimedes AI

Automated LeanIX Population

- SAP Advisory Architecture - May 2026



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# Agenda

- 01 The challenge — manual LeanIX population
- 02 What is Archimedes AI
- 03 How it works — 5-step pipeline + SAP Help contrast
- 04 Real case — Acciona
- 05 Value for the Advisory Architect
- 06 Getting started

THE CHALLENGE

# Manual LeanIX Population

# Manual LeanIX Population: A Costly Process

- 2-3 days of manual work per client engagement to populate LeanIX
- Multiple disconnected sources: OnPrem/Cloud Excel, requirements, PDFs, architecture diagrams
- BC coverage depends on individual knowledge of the RBA catalog (756 BCs, 22 domains)
- Application-to-RSA mapping is manual, subjective, and error-prone
- Not scalable — depends on who knows the SAP catalog in the team
- LeanIX stays empty or incomplete until late in the engagement



THE SOLUTION

# What is Archimedes AI

# Archimedes AI – Overview

## WHAT IT DOES

- AI-powered orchestrator: client inputs → LeanIX import files
- Covers full EA cycle: AS-IS Baseline + TO-BE Target
- Runs locally – client data stays on your machine
- Output directly importable into LeanIX via GraphQL

## THE ENGINE

- Claude API (claude-sonnet-4-6)
- SAP RBA: 756 Business Capabilities, 22 domains
- SAP RSA: 324 canonical SAP products
- All AI calls routed via SAP Hyperspace AI gateway

```
python3 run.py pipeline --client <name>
```



# Inputs and Outputs

| INPUT TYPE            | SOURCE                   | WHAT ARCHIMEDES EXTRACTS                         |
|-----------------------|--------------------------|--------------------------------------------------|
| OnPrem Systems Excel  | Client-provided          | Applications + hosting tags (Baseline AS-IS)     |
| Cloud Systems Excel   | Client-provided          | Cloud apps + solution area tags (Baseline AS-IS) |
| Requirements Excel    | Client-provided          | BCs (RBA mapping) + SAP apps (RSA mapping)       |
| PDF documents         | Proposals, RFPs, reports | Fact sheets: Applications, BCs, Initiatives      |
| Architecture diagrams | PNG / JPG images         | Applications and IT components (Claude Vision)   |
| baseline.xlsx         | → LeanIX                 | AS-IS applications with Baseline tags            |
| target_leanix.xlsx    | → LeanIX                 | TO-BE: Application, BC, Initiative, ITComponent  |

— HOW IT WORKS

# The 5-Step Pipeline





# Pipeline – Steps 0 to 2

## 0

### Catalog Check

Verifies RBA/RSA catalog version and date.  
Offers update from Value Experience Hub.  
Ensures the knowledge base is current before processing.

## 1

### Baseline AS-IS

Reads OnPrem + Cloud Systems Excel →  
generates **baseline.xlsx** with  
**Baseline;OnPremise** / **Baseline;Cloud**  
tags per application.

## 2

### Requirements TO-BE

Reads requirements Excel → Claude maps  
each row to RBA Business Capabilities and  
RSA canonical products. Also processes  
additional PDFs for supplementary fact  
sheets.



## Pipeline – Steps 3 to 5

### 3

#### Images & Diagrams

Reads architecture diagrams (.png / .jpg). Claude Vision extracts applications and IT components directly from visual diagrams – no manual transcription needed.

### 4

#### Output Generation

Produces multi-sheet LeanIX Excel: **Application · BusinessCapability · Initiative · ITComponent · ReadMe**. Each sheet is import-ready.

### 5

#### LeanIX Import

Optional GraphQL push in the correct import order: **BC → ITC → App → Initiative**. Requires **LEANIX\_API\_TOKEN** in **.env**.



# Industry Reference Architecture – Whitespace Analysis

## What it does

- Fetches SAP-recommended products for the client's industry from the **SAP Business Accelerator Hub** (public OData API)
- Compares reference products against both **Baseline** and **Target** LeanIX Excels
- Shows gaps (whitespace) as a checklist – products recommended by SAP not yet covered
- Adds selected products to the Target Excel with `tag="Target Reference", lifecycle="plan"`
- Derives **BC + ITC relations** per product via Claude API and RBA catalog

## Wizard – Step 8

- Industry selector: 21 SAP industries (EC&O, Retail, Automotive, Utilities...)
- Load button → fetches reference → shows gap list with checkboxes
- *Already covered* products (Baseline or Target) clearly marked
- Select all / individual checkbox → Apply → products written to Target Excel
- Import order preserved: BC → App (with `relApplicationToBusinessCapability + relApplicationToITComponent`)

# SAP Help Portal Contrast

## What it does

- After enrichment, optionally validates each RSA product mapping against **help.sap.com**
- Searches official SAP documentation for each requirement's assigned product
- Flags mappings with no direct documentation confirmation
- Generates a [help\\_contrast\\_report.json](#) with per-requirement results

## How to use

- **CLI:** prompted automatically after Step 2 enrichment — answer **s** to activate
- **Wizard:** checkbox appears after requirements are processed
- Completely optional — pipeline continues either way
- Results: ✓ Validated · ⚠ Review · — No RSA assigned

# LeanIX API Enhancements

## Reliability & Performance

- **Retry 429 with backoff** – GraphQL calls auto-retry on rate-limit with exponential backoff (reads [Retry-After](#) header); also retries on network errors
- **Batch GraphQL (x20)** – Fact sheet lookups use aliased multi-query calls (up to 20 per round-trip), reducing API calls from N to  $\lceil N/20 \rceil$
- **ITComponent push** – ITC sheet from Excel is now upserted into LeanIX before Applications (correct import order: BC → ITC → App → Initiative)

## Catalog Intelligence & TO-BE Preview

- **App & ITC Catalog linking (fix 2026-06-17)** – Payload correcto via [GET /reference-data/v1/openapi.json](#): campo `sourceFactSheetIdentifier: {id, externalId}` (objeto anidado). Payload plano anterior (`sourceFactSheetId`) retornaba `null` y status `ignored`. Flujo: [POST batch-links](#) → resolver UUID + externalId → [PUT /links](#).
- **ITC Catalog linking** – ITComponents auto-linked to official LeanIX catalog entries (`lx_ITC_XXXXXX`) via [/source/ttls/](#) API (mirrors existing App linking)
- **Enrich from catalog** – After VERYHIGH match, copies [hostingType](#) and [description](#) from catalog fact sheet to workspace Application
- **Impacts API – Preview TO-BE** – New wizard button calls [POST /impacts/v1/projections](#): shows projected field changes for each Application given all active Transformations and a target date



# RBA / RSA Catalogs – Always Up to Date

From Manual Export to Live API

- **Before:** catalogs updated manually via Excel export from the SAP Value Experience Hub – error-prone, monthly at best
- **Now:** one click in the wizard calls the **MXP MCP server** (SAP Model Experience Platform) and regenerates both JSONs in seconds
- **Auth:** automatic via SAP SSO (I519409 identity) – no token to manage
- **Scope:** [dc676573](#) worksphere – SAP Reference Architecture

What Changed – May 2026

- **RBA:** 1,141 BCs across 29 domains & 148 areas (*was 756*)
- **RSA:** 515 SAP products across 16 domains (*was 324*)
- **Wizard Step 1** shows last-update age and a “Refresh from MXP” button – on-demand, not on every run
- [/update-catalogs](#) Claude Code slash command for direct MCP-based refresh without a REST token

— REAL CASE

# Acciona

# Acciona – Pipeline Results

**37** apps

BASELINE TOTAL

**462** rows

REQUIREMENTS PROCESSED

**20** BCs

BUSINESS CAPABILITIES (RBA)

**9** init.

INITIATIVES BY PROCESS

| PHASE          | METRIC                                  | RESULT                       |
|----------------|-----------------------------------------|------------------------------|
| Baseline AS-IS | On-Premise systems                      | 14 apps (Baseline;OnPremise) |
| Baseline AS-IS | Cloud systems                           | 23 apps (Baseline;Cloud)     |
| Target TO-BE   | SAP applications mapped (RSA)           | 7 canonical SAP apps         |
| Target TO-BE   | Business Capabilities with parent chain | 20 leaf BCs                  |
| Target TO-BE   | IT Components derived                   | 8 ITCs                       |





## Acciona – What This Means

- Full LeanIX population for a real client engagement – from baseline to target in one run
- 462 requirements processed and mapped to RBA/RSA in minutes, not days
- BC hierarchy automatically built from RBA: leaf nodes + full parent chain
- Initiatives grouped by process (col A), not by individual requirement – ready for roadmap
- IT Components auto-derived from the app-to-ITC static map (S/4HANA, SAC, IBP, Ariba...)
- Output imported into LeanIX directly – Advisory Architect can start working on day 1

THE VALUE

# Advisory Architect Impact

# Without vs. With Archimedes AI

| ACTIVITY              | WITHOUT ARCHIMEDES                 | WITH ARCHIMEDES                           |
|-----------------------|------------------------------------|-------------------------------------------|
| LeanIX population     | ✗ 2-3 days manual work             | ✓ Minutes (automated pipeline)            |
| BC coverage (RBA)     | ✗ Depends on individual knowledge  | ✓ Guaranteed – 756 BCs, 22 domains        |
| App-to-RSA mapping    | ✗ Manual, subjective               | ✓ Canonical resolution (324 SAP products) |
| Multi-source handling | ✗ Each source processed separately | ✓ One orchestrator, all sources           |
| PDF / diagram input   | ✗ Manual extraction                | ✓ Claude API vision + NLP                 |
| LeanIX import         | ✗ Manual upload + mapping          | ✓ Direct GraphQL push                     |
| Scalability           | ✗ Depends on catalog expert        | ✓ Any Advisory Architect, any client      |

# The Advisory Architect's Time – Refocused

BEFORE ARCHIMEDES

## Mechanical Work

- Manual data entry into LeanIX
- Catalog lookup (RBA/RSA)
- Spreadsheet formatting
- Waiting for complete data



AFTER ARCHIMEDES

## Strategic Work

- Strategic validation of the model
- Gap analysis and recommendations
- Client-facing decisions
- EA ready on day 1

HANDS ON

# Getting Started

# Running Archimedes AI

**1**

## Prerequisites

Python 3.10+ · [ANTHROPIC\\_API\\_KEY](#) · optional: [LEANIX\\_API\\_TOKEN](#)

**2**

## Repository

```
git clone https://github.tools.sap/I519409/archimedes_ai.git
```

**3**

## Install dependencies

```
pip install -r requirements.txt
```

**4**

## Run the pipeline

```
python3 run.py pipeline --client <client_name>
```

• Interactive – prompts for each input file

• Output: output/<client>/baseline.xlsx + target\_leanix.xlsx

• Idempotent – safe to re-run, no LeanIX duplicates

• Resume support – enrichment checkpoints every 10 reqs

• PDF & diagram fact sheets merged automatically into Excel



# Multi-Workspace Support – Step 0

## What Changed

- **Before:** credentials fixed in `.env` – one workspace per deployment
- **Now:** Step 0 (Client & Workspace) shows a **workspace dropdown** populated from `workspaces.json`
- User selects a saved workspace → *Validate* → green badge confirms connection + app count
- Or clicks “+ Create new workspace...” → inline form (Name, Base URL, API Token) → *Save & Validate*
- All subsequent pipeline steps use that workspace's credentials – not `.env`

## Technical Design

- `workspaces.json` – project-root file; tokens in plaintext (same model as `.env`); git-ignored
- `GET /api/workspaces` – returns name + URL only (no tokens)
- `POST /api/workspaces/validate` – validates via LeanIX GraphQL, returns app count
- `_leanix_creds(sess)` – session-first credential lookup; falls back to `.env` (backwards-compatible)
- Header badge  **<workspace>** always visible after validation – no ambiguity